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Multi-scale effects of impoundments on genetic structure of creek chub (*Semotilus atromaculatus*) in the Kansas River basin

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SUMMARY

1. Habitat fragmentation has been implicated as a primary cause for the ongoing erosion of global biodiversity, yet our understanding of the consequences in lotic systems is limited for many species and regions. Because of harsh environmental conditions that select for high colonisation rates, prairie stream fishes may be particularly vulnerable to the effects of fragmentation. Hence, there is urgent need for broader understanding of fragmentation in prairie streams such that meaningful conservation strategies can be developed. Further, examination at large spatial scales, including multiple impoundments and un-impounded catchments, will help identify the spatial extent of species movement through the landscape.

2. Our study used data from 10 microsatellite loci to describe the genetic structure of creek chub (*Semotilus atromaculatus*) populations across four catchments (three impounded and one un-impounded) in the Kansas River basin. We investigated whether genetic diversity was eroded in response to habitat fragmentation imposed by reservoirs and whether intervening lentic habitat increased resistance to dispersal among sites within a catchment.

3. Our analyses revealed that genetic diversity estimates were consistent with large populations regardless of the location of the sampled tributaries, and there was little evidence of recent population reductions. Nevertheless, we found a high degree of spatial genetic structure, suggesting that catchments comprise a set of isolated genetic units and that sample sites within catchments are subdivided into groups largely defined by intervening habitat type. Our data therefore suggest that lentic habitat is a barrier to dispersal among tributaries, thus reducing the opportunity for genetic rescue of populations in tributaries draining into reservoirs. Isolation by a reservoir, however, may not be immediately deleterious if the isolated tributary basin supports a large population.

Keywords: creek chub, conservation genetics, connectivity, waterway impoundment, landscape genetics

Introduction

Ecological and evolutionary dynamics of populations are linked to the area and spatial arrangement of habitat patches on a landscape (MacArthur & Wilson, 1967; Hanski & Gilpin, 1991), and natural landscape features influence the size, shape and connectivity of habitat patches (Castric, Bonney & Bernatchez, 2001; Blaum & Wichmann, 2007, Croteau *et al*., 2007). Anthropogenically induced fragmentation of habitat eliminates critical dispersal pathways, altering dispersal behaviour (e.g. Stow *et al*., 2001) and isolating populations (Yamamoto *et al*., 2004; Wofford, Gresswell & Banks, 2005; Bergl & Vigilant, 2007). The disruption of migration corridors reduces gene flow, leading to stochastic genetic differentiation among populations (Nei, Maruyama & Chakraborty, 1975) and eventually resulting in the reduction in total genetic variation within a taxon (e.g. Epps *et al*., 2005; Allendorf & Luikart, 2007). Habitat fragmentation has therefore been implicated as a primary cause underlying the global

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erosion of genetic diversity (Allendorf & Luikart, 2007) and ultimately biodiversity (Primack, 1993).

Spatial processes in streams and rivers are functionally different than those of terrestrial systems because their linear nature and hierarchical branching structure constrain movement of organisms (McGlashan, Hughes & Bunn, 2001; Campbell Grant *et al*., 2007). Fragmentation of lotic habitats disrupts dispersal within stream networks and can bias the direction of movement if barriers can be traversed downstream but not upstream (e.g. Nehlsen, Williams & Lichatowich, 1991; Ruban, 1997). These disruptions may result in reduced local population size and increased extinction probability if preferred habitat becomes unavailable or inaccessible. Stream organisms, therefore, may be particularly sensitive to changes in connectivity that can isolate demes. Although dams, irrigation diversions and other waterway alterations are common features in contemporary aquatic systems (Benke, 1990), the genetic effects of fragmentation in running water habitats has been restricted to a limited number of taxa, many of which occur in mainstem habitats (e.g. Castric *et al*., 2001; Meldgaard, Neilsen & Loeschcke, 2003; Neville, Dunham & Peacock, 2006; Bessert & Orti, 2007; Leclerc *et al*., 2008; Meeuwig *et al*., 2010). Nevertheless, recent evidence suggests that impacts of habitat fragmentation on genetic structure of stream fishes are likely to depend on ecological traits of those species. For example, Lamphere & Blum (2012) found limited dispersal of a benthic dwelling fish over a relatively small distance (<5 km). Alternatively, Aló & Turner (2005) found that a large river cyprinid required dispersal across a river section longer than 100 km. Hence, there is an urgent need to understand the genetic consequences and potentially infer the demographic consequences of fragmentation in different lotic habitats because a clear understanding of patterns of genetic diversity is of central importance to developing conservation strategies that encourage the persistence of aquatic organisms in changing landscapes (Ryman, Utter & Laikre, 1995; Tibbets & Dowling, 1996; Aló & Turner, 2005).

Prairie stream fishes may be particularly vulnerable to fragmentation because harsh environmental conditions (e.g. drought) result in increased extinction probability (Dodds & Oakes, 2004), and impoundment of prairie streams restricts recolonisation (e.g. Winston, Taylor & Pigg, 1991). Because of these stochastic environmental conditions, prairie stream fishes have life-history traits that maximise persistence in the face of increased extinction probability (e.g. increased colonisation and long-distance dispersal). Nevertheless, several recent studies have evaluated the consequence of reservoir fragmentation on prairie stream fishes. Falke & Gido (2006) showed distinct communities in tributaries that flow into reservoirs in contrast to those flowing into mainstem rivers. Matthews & Marsh-Matthews (2007) reported the extirpation of a common prairie stream fish in tributaries that directly connected to a reservoir, but not in populations upstream of the reservoir. Franssen (2012) found decreased genetic connectivity among tributary populations that have to move through reservoirs, in contrast to genetic isolation by distance among tributary populations that have to move through a mainstem stream. Similarly, Skalski *et al.* (2008) observed a pattern of eroded genetic diversity and disrupted connectivity for creek chub in North Carolina (USA), suggesting that reservoirs are isolating mechanisms for fishes in prairie and non-prairie streams alike. Because of the particularly harsh conditions and higher extinction probabilities typical of prairie streams, responses to fragmentation might be variable across species, spatial scales and time periods, thus requiring further multi-scale investigations to identify key factors influencing population viability in these fragmented systems. Franssen (2012) and Skalski *et al.* (2008) examined the effects of a single impoundment on connectivity within a catchment, but examination at a more extensive spatial scale, including multiple impoundments and un-impounded catchments, could provide useful insights into the general effect of impoundments on population connectivity.

Our study describes the genetic structure of creek chub (*Semotilus atromaculatus:* a headwater minnow) across four catchments (three impounded and one un-impounded) in the Kansas River basin. We address two questions: (i) Is genetic diversity eroded in creek chub populations occupying tributaries upstream of, or directly connected to, a reservoir as compared to tributary populations downstream of reservoirs or in un-impounded catchments? and (ii) Does intervening lake habitat type (lentic versus lotic) increase resistance to dispersal among sites within a catchment? To address our second question, we evaluated three movement hypotheses described in the conceptual framework developed by Thornbrugh & Gido (2010). Their limited exchange model predicts that dispersal is restricted to habitats within a tributary or mainstem reach, which would result in a pattern of high genetic isolation among populations in separate (i.e. isolated) tributaries. In contrast, the confluence exchange and network dispersal models would predict a pattern of genetic differentiation consistent with isolation by distance where the extent of genetic isolation will be driven by the dispersal capability of the focal taxon.

Methods

Study system

We conducted our study of genetic structure on creek chub by collecting tissue samples from individuals captured at multiple sites in four catchments of the Kansas River basin (Kansas, USA; Fig. 1, Table 1). Creek chub are widespread and abundant in the eastern and north-central United States and are categorised as a small-stream species (Jenkins & Burkhead, 1993) associated with headwater streams in the Great Plains (Cross & Collins, 1995). Available information from our study area in Kansas suggests that creek chub are relatively rare in larger streams and rivers (Eitzmann, 2008; Paukert, Eitzmann & Gerkin, 2009). Nevertheless, their presence suggests that this species readily uses these network components as dispersal corridors. Hence, creek chub provides a convenient and widely applicable study system for generalising the effects of anthropogenic barriers on genetic diversity within and connectivity among populations of fishes in headwater streams.

Clinton Lake, Perry Lake and Tuttle Creek Lake impound the Wakarusa River, Delaware River and Big Blue River and have been in place since 1977, 1969 and 1959, respectively. These reservoirs are operated by the U.S. Army Corps of Engineers and are primarily used for water storage, flood control and recreation. Although detailed demographic data for creek chub are scarce, available age data suggest they have a 2- to 5-year generation time (Jenkins & Burkhead, 1993; Quist & Guy, 2001). Thus, these impoundments have potentially been affecting creek chub population structure in the Kansas River basin for a maximum of about 15–24 generations.

Sample collection, DNA extraction and microsatellite amplification

We collected 9–37 creek chub from 25 tributaries flowing into the three impoundments (Clinton Lake, n = 14; Perry Lake, n = 6; Tuttle Creek Lake, n = 5) and nine tributaries flowing into an un-impounded drainage (Mill Creek) by backpack electrofishing during the summers of 2006 and 2008 (Fig. 1). To collect tissue, we clipped a portion of the anal fin from each individual and preserved the sample in 70% ethanol for transport to the laboratory. Genomic DNA was extracted by tissue digestion in cell lysis buffer (10 mM Tris, 100 mM EDTA, 2% SDS, pH = 8.0) with

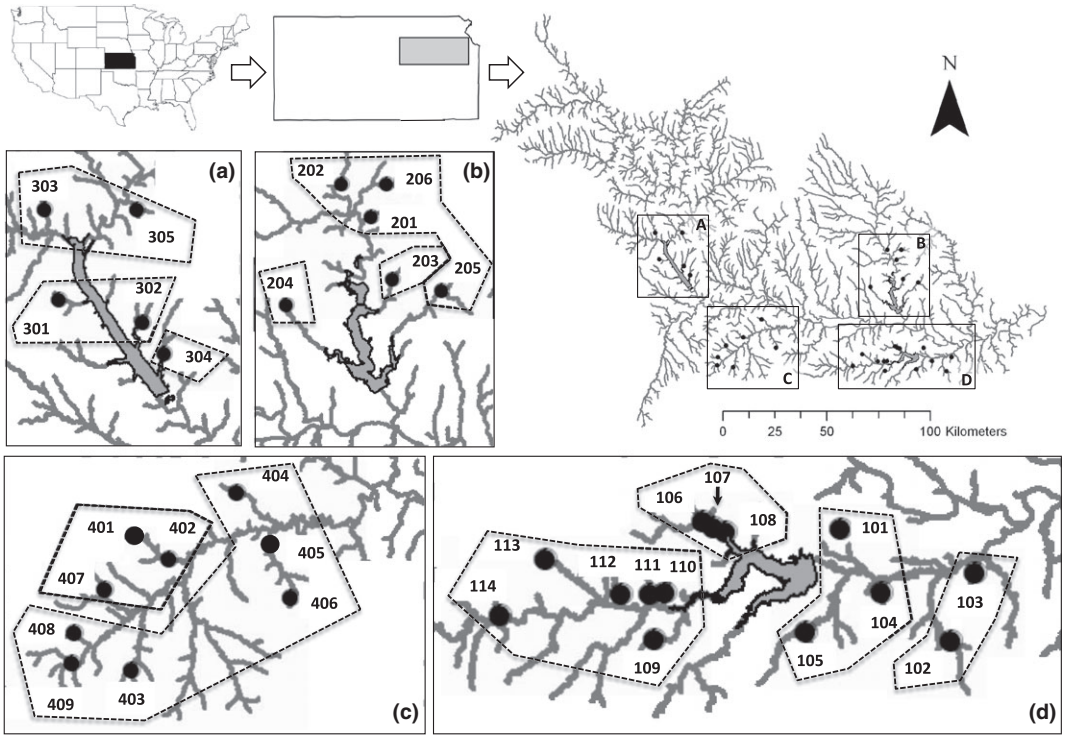

Fig. 1 Our study of creek chub (*Semotilus atromaculatus*) genetic diversity comprises 34 geographical locations from four catchments in the Kansas River basin (Kansas, USA): (a) Tuttle Creek Reservoir (in place since 1959) impounds the Big Blue River, (b) Perry Lake (in place since 1969) impounds the Delaware River, (c) Mill Creek (un-impounded) and (d) Clinton Lake (in place since 1977) impounds the Wakarusa River. Site numbers correspond to those presented in Table 1. Polygons enclose those sites that were included in the same genetic cluster and correspond with the results from STRUCTURE presented in Fig. 2; Site 204 (Panel b) clusters with Sites 102 and 103 (Panel d).

Table 1. Genetic diversity estimates (Allelic Richness, Â; Gene Diversity, H_E, and Inbreeding, F_{IS}), the number of individuals sampled (*n*) and the mean number of individuals genotyped (*G*) for 34 sampling locations in tributaries from four catchments in the Kansas River basin (Kansas, USA)

Catchment	Site	Latitude, Longitude	Â	H_E	F_{IS}	*n*	*G*
Clinton Lake	101	38.95860ºN, 095.30002ºW	3.7	0.61	0.069	24	23.8
	102	38.85363ºN, 095.17710ºW	4.6	0.69	0.023	23	22.9
	103	38.91338ºN, 095.14615ºW	4.2	0.66	0.119	23	23
	104	38.89893ºN, 095.25560ºW	4.1	0.65	0.051	25	24.8
	105	38.86683ºN, 095.34480ºW	4.1	0.62	−0.041	22	21.6
	106	38.96970ºN, 095.45923ºW	3.4	0.59	0.014	23	23
	107	38.96432ºN, 095.44665ºW	3.6	0.62	−0.060	13	11.5
	108	38.95947ºN, 095.43228ºW	3.5	0.60	0.127	31	30.6
	109	38.86578ºN, 095.51928ºW	3.7	0.62	−0.050	25	25
	110	38.90788ºN, 095.50490ºW	4.2	0.68	−0.011	24	24
	111	38.90712ºN, 095.52332ºW	3.8	0.65	−0.079	20	18.1
	112	38.90737ºN, 095.55625ºW	4.2	0.67	−0.055	27	27
	113	38.94203ºN, 095.64235ºW	3.9	0.62	0.060	25	25
	114	38.89298ºN, 095.69778ºW	4.4	0.70	0.011	23	23
Perry Lake	201	39.34633ºN, 095.43025ºW	4.6	0.71	−0.011	25	23.3
	202	39.39052ºN, 095.47955ºW	4.6	0.73	0.032	18	15.8
	203	39.26112ºN, 095.39887ºW	4.0	0.63	0.008	25	24
	204	39.23282ºN, 095.58355ºW	3.9	0.64	0.050	17	14.7
	205	39.24423ºN, 095.31442ºW	4.1	0.66	0.004	25	24.9
	206	39.38847ºN, 095.40165ºW	4.4	0.68	0.066	24	21.7
Tuttle Creek Lake	301	39.38137ºN, 096.75947ºW	4.9	0.73	0.094	27	26.9
	302	39.34994ºN, 096.62282ºW	4.4	0.67	0.041	32	30.8
	303	39.49750ºN, 096.77722ºW	4.3	0.67	0.006	37	36.5
	304	39.30968ºN, 096.58905ºW	4.2	0.67	0.004	32	31.4
	305	39.49333ºN, 096.62583ºW	4.4	0.67	0.008	31	30.8
Mill Creek	401	39.03237ºN, 096.30075ºW	4.2	0.63	0.008	32	36.5
	402	39.06144ºN, 096.35672ºW	4.7	0.72	0.042	16	16
	403	38.90437ºN, 096.36208ºW	4.3	0.67	0.082	16	16
	404	39.10783ºN, 096.19698ºW	4.3	0.66	0.010	30	30
	405	39.05997ºN, 096.14798ºW	4.5	0.63	0.040	9	9
	406	38.98313ºN, 096.12083ºW	4.4	0.66	0.060	33	32.5
	407	38.99888ºN, 096.39968ºW	4.3	0.63	0.041	29	28.8
	408	38.91377ºN, 096.45095ºW	4.5	0.67	0.129	29	28.7
	409	38.94933ºN, 096.44562ºW	4.0	0.63	0.055	22	22

proteinase K (Life Technologies, Inc., Grand Island, NY, USA), treatment with RNase A (Qiagen, Inc., Valencia, CA, USA), ammonium acetate precipitation of proteins and alcohol precipitation of DNA before hydration in TLE buffer (10 mM Tris, 0.1 mM EDTA, pH = 8.0). We quantified DNA concentration in our isolates using an ND-1000 spectrophotometer (Thermo Scientific, Wilmington, DE, USA) and diluted each sample to a concentration of about 20 ng μL^{-1} for use as template in polymerase chain reaction (PCR). Our genetic data set comprised the genotypes of 9–37 individuals from each sampling location amplified at 16 microsatellite loci using four multiplexed PCRs (Table S1), fragment detection on an ABI 3130XL Genetic Analyser (Life Technologies, Inc., Grand Island, NY, USA) and fragment sizing using GENEMARKER 1.80 software (SoftGenetics®, State College, PA, USA) calibrated with the LIZ(-250) size standard.

Of the 16 microsatellite markers targeted for analysis, one (*Seat206*) amplified inconsistently and was not scored or included in any analyses. Five additional markers consistently deviated from Hardy–Weinberg proportions (MCMC permutation test using 1000 batches of 10 000 dememorisation steps followed by 10 000 iterations in GENEPOP v3.4; Raymond & Rousset, 1995) and demonstrated levels of heterozygote deficiency consistent with the presence of null alleles. Removal of these six markers resulted in a 10-locus data set demonstrating no consistent deviations from Hardy–Weinberg proportions and no evidence of linkage disequilibrium (MCMC permutation test using 1000

batches of 10 000 dememorisation steps followed by 10 000 iterations in GENEPOP v3.4) over all sites following correction for multiple inferences using a stepwise Bonferroni procedure (Sokal & Rohlf, 1995). All subsequent results were informed by the 10-locus data set.

Data analysis

Effects of impoundment on genetic diversity. For those loci in Hardy–Weinberg and linkage equilibrium, we estimated rarefied allelic richness (Â; Petit & El Mousadik, 1998; Leberg, 2002), gene diversity (H_E) and the inbreeding coefficient (F_{IS}) using FSTAT v2.9.3 (Goudet, 1995). We characterised each site sampled by taking the average over loci for each estimate of genetic diversity (Table 1). To test the effects of impoundment on creek chub genetic diversity, we used analysis of variance (ANOVA) to compare patterns of allelic richness, gene diversity and inbreeding in our study area. For each estimate, we used catchment as a main effect in a single-factor model and, when a significant system-level effect was detected, drew contrasts (i) between impounded (i.e. tributaries directly connected to an impoundment or upstream of an impoundment) and un-impounded (downstream of an impoundment or in an un-impounded catchment) sample sites, (ii) among impounded sample sites in different systems and (iii) between un-impounded sample sites. For the purpose of these analyses, the Clinton Lake sites were divided into two systems: impounded (those sample sites upstream of the impoundment) and un-impounded (those sample sites downstream of the impoundment). This same general framework was used to analyse patterns of differentiation in three response variables, namely mean allelic richness, mean gene diversity and mean signal of inbreeding. We also assessed the probability that each of the sampled sites in our study area had undergone genetic bottlenecking using the BOTTLENECK software package (Cornuet & Luikart, 1996).

Spatial genetic structure. We estimated genetic differentiation among sample sites in four ways. First, we used GENEPOP v3.4 (Raymond & Rousset, 1995; Rousset, 2008) to test for genotypic differentiation between each pair of sites by implementing an MCMC permutation test (same parameters as for Hardy–Weinberg and Linkage equilibrium) followed by the stepwise Bonferroni procedure (Sokal & Rohlf, 1995). Second, we assessed the overall hierarchical partitioning of genetic variation using analysis of molecular variance (AMOVA; Excoffier, Smouse & Quattro, 1992) implemented in GENODIVE v3.1 (Meirmans & Van Tienderen, 2004). Third, we examined spatial genetic structure among catchments using the clustering algorithm STRUCTURE (v2.3; Pritchard, Stephens & Donnelly, 2000). We modelled the genetic structure in our study area using an empirically determined allele frequencies parameter ($\lambda = 0.738$), under an admixture model with correlated allele frequencies and informative locations (LOCPRIOR = 1). Our objective was to use our genetic data to differentiate between a set of categorical classifications of our sample sites determined by intervening habitat type and spatial arrangement of sampled sites. Sample sites were assigned to a location based upon the catchment within which each was located and by the type of habitat separating any pair of sample sites; those sites within a catchment that were connected by lotic corridors were indexed as a single location, and those sites separated by lentic habitat as separate populations (i.e. populations are thought to be delimited by exchange of migrants; Waples & Gaggiotti, 2006). We allowed k to vary from 4 to 12, and our strategy resulted in 11 potentially informative site groups (Fig. 1; Site 204 clusters with Sites 102 and 103). Fourth, we calculated the likelihood ratio genetic distance (D_{LR}, Paetkau *et al.*, 1997) and examined this statistic as a function of waterway distance [the distance along the shortest path through the water body connecting each sampling location pair calculated using ArcGIS (ESRI, 2011)] to summarise spatial patterns of pairwise population differentiation over the entire study area. We used the statistical package R (R Development Core Team, 2009) to implement 10 000 iterations of Mantel's permutation test (Mantel, 1967) to determine the pattern of genetic isolation with respect to geographical distance [isolation by distance (IBD)].

Resistance to movement associated with impoundments. To understand whether impoundments increase resistance to dispersal, we first examined genetic distance (D_{LR}) as a function of waterway distance and intervening habitat type using a nonparametric ANCOVA. We then examined patterns of isolation by distance for the subsets of populations (regardless of catchment) separated by lotic habitat (e.g. Site 101 and Site 102, Fig. 1d), lentic habitat (e.g. Site 106 and Site 109, Fig. 1d), lentic habitat and a dam (those populations in the Wakarusa river drainage downstream of the Clinton Lake Reservoir; e.g. Site 101 and Site 109, Fig. 1d) and sites separated into different catchments (i.e. those separated by lotic and lentic habitat with at least one dam in the putative migration corridor; e.g. Site 101 and Site 201, Fig. 1). As before, we used an algorithm written in R (R Development Core Team, 2009; code by SPH) to implement 10 000 iterations of Mantel's

permutation test (Mantel, 1967) to determine the pattern of genetic isolation with respect to geographical distance (IBD) in each subset of the data.

To augment our examination of IBD patterns in this catchment, we applied a nonparametric ANCOVA to determine the effect of intervening habitat type on genetic differentiation in the Wakarusa River catchment because we sampled population pairs upstream of and downstream of the Clinton Lake Dam (Fig. 1), thus allowing a direct contrast of isolation-by-distance patterns associated with intervening habitat type. We used an algorithm written in R (R Development Core Team, 2009; code by SPH) and implemented 10 000 permutations of the nonparametric ANCOVA by shuffling intervening habitat type to maintain the pairwise relationship between waterway distance and D_{LR} estimates. We assessed the significance of the effect of intervening habitat by calculating the difference in mean D_{LR} between lotic-separated and lentic-separated sites and comparing the observed difference to the distribution of permuted differences.

Finally, we used a Bayesian clustering algorithm implemented in BayesAss v3.0 (Wilson & Rannala, 2003) to detect the signature of recent movement among sampled sites within a catchment. We used five replicate runs (each with a different seed) with a burn-in of 10^6 iterations, sampling for 10^7 iterations and data collection every 10^3 steps during sampling. Values for the migration (m), allele frequencies (a) and inbreeding (f) switching proposals were empirically determined for each catchment such that acceptance rates were between 20% and 40%, as suggested by Rannala (2007). Within each catchment, we categorised population pairs into those separated by lotic habitat, those separated by lentic habitat and those separated by lentic habitat and a dam (Clinton Lake/Wakarusa River system only). We estimated the mean signature of movement within each category by taking the sum of the off-diagonal values, which represents the proportion of the individuals sampled at each site thought to be of migrant ancestry.

Results

Effects of impoundment on genetic diversity

Estimates of genetic diversity taken from the 34 sample sites included in this study suggest that creek chub populations surrounding individual sample sites are large. We observed an average of 4.8–8.8 alleles per locus, with rarefied estimates ($\hat{A}$; $n = 5$ diploid individuals per site) ranging from 3.5 to 4.9 (Table 1). Estimates of mean gene diversity (H_E) ranged from 0.59 to 0.73, and mean signal of inbreeding (F_{IS}) was low across the study area, ranging from −0.10 to 0.13 (Table 1). Neither average H_E nor F_{IS} varied significantly among catchments ($P > 0.05$; Table 2). Mean $\hat{A}$, however, showed significant variation among catchments ($P < 0.006$) driven primarily by variation among impounded catchments ($P < 0.004$) rather than between impounded and un-impounded groups of populations ($P > 0.05$) or between the two groups of populations in un-impounded catchments ($P > 0.05$; Table 2). Testing for population-level heterozygote excess revealed that the majority of our sample sites had not

Table 2 Results from analysis of variance on three measures of genetic diversity estimated for 34 sample sites distributed among five systems in the Kansas River basin: Clinton Lake, Wakarusa River — for the purpose of these analyses, the Clinton Lake catchment was divided into two systems: impounded (those sample sites upstream of the impoundment) and un-impounded (those sample sites downstream of the impoundment) — Perry Lake, Tuttle Creek Lake and Mill Creek. Contrasts between impounded (directly connected to an impoundment or upstream of an impoundment) and un-impounded (downstream of an impoundment or in an un-impounded catchment) sample sites, among impounded sample sites in different systems and between un-impounded sample sites were carried out when a significant system effect was detected

Response variable	Source of variation	d.f.	SS	MS	F	Pr ($F \mid H_o$)
Gene diversity (H_E)	System	4	0.0089	0.0022	1.9551	0.1280
	Error	29	0.0329	0.0011		
	Total	33				
Inbreeding (F_{IS})	System	4	0.0090	0.0023	0.8690	0.4943
	Error	29	0.0752	0.0026		
	Total	33				
Allelic richness ($\hat{A}$)	System	4	1.5335	0.3834	4.6383	**0.0051**
	Impounded vs. Un-impounded	1	0.2199	0.2199	2.6603	0.1137
	Among Impounded	2	1.1461	0.5731	6.9335	**0.0035**
	Between Un-impounded	1	0.1674	0.1674	2.0259	0.1653
	Error	29	2.3969	0.0827		
	Total	33				

Table 3 AMOVA results showing significant partitioning of genetic variation among catchments, populations within catchments and individuals within populations

Source of variation	% Variation	φ-Value	SE φ	P-value
Among catchments (F_{CT})	3.1	0.031	0.006	< 0.001
Among population within catchment (F_{SC})	6.3	0.065	0.010	< 0.001
Among individual within population (F_{IS})	2.3	0.026	0.011	< 0.001
Within individual (F_{IT})	88.3	0.117	0.018	

undergone recent bottlenecks (BOTTLENECK v1.2.02; Cornuet & Luikart, 1996). Two sample sites upstream of Perry Lake (Site 202 and Site 206), however, indicated significant signal for recent genetic bottleneck (Wilcoxon Test, $P < 0.05$).

Spatial genetic structure

Evidence of a high degree of spatial genetic structure among the sample sites in our study area arose from three lines of evidence. First, genotypic differentiation was significant between all sites and across all loci following correction for multiple inferences (data not shown). Second, AMOVA indicated significant spatial partitioning of genetic variation coupled with high levels of genetic diversity within populations. The largest proportion of genetic variation was contained within individuals (*c.* 88%), and significant partitioning was indicated at all hierarchical levels (Table 3).

Analysis of genetic structure revealed 11 clusters that generally corresponded with catchment and intervening habitat type (Fig. 2). Sample sites surrounding Clinton Lake were partitioned into four main clusters largely corresponding to intervening habitat type (Figs 1 and 2). One cluster corresponded to the sample sites separated by lotic habitat and located in the north arm of the impoundment. A second cluster corresponded to the sample sites separated by lotic habitat and located along the Wakarusa River west of the impoundment. The third and fourth clusters are nested within the sample sites separated by lotic habitat and located along the Wakarusa River east of the impoundment and downstream of the dam. Sample sites surrounding Perry Lake were partitioned into three clusters largely corresponding to the type of intervening habitat: sample sites separated by lotic habitat north of the lake, a sample site east of the lake and a sample site west of the lake (Figs 1 and 2). The sample site west of the lake (Site 204) had allele frequencies in common with two sites from the Wakarusa River cluster east of Clinton Lake. Sample sites surrounding Tuttle

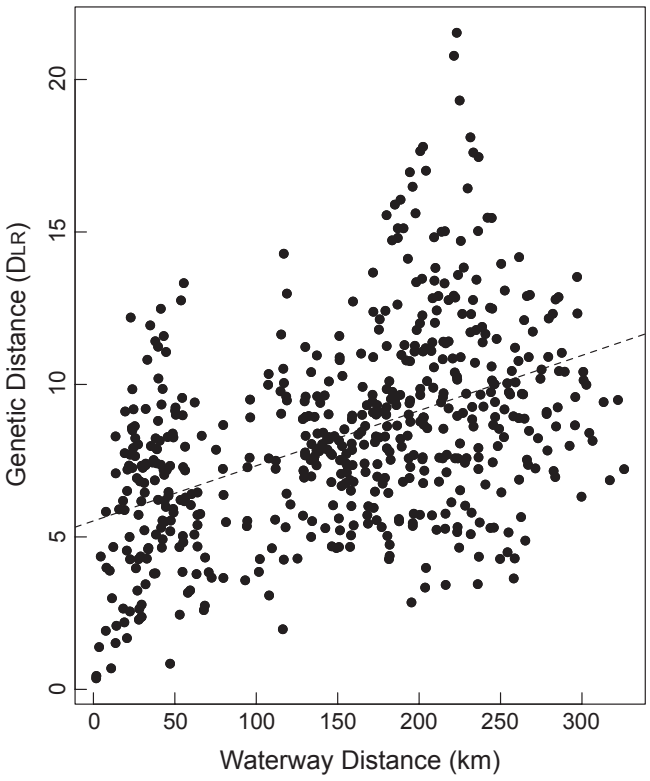

Fig. 3 Relationship between pairwise waterway distance (km) and pairwise genetic distance (D_{LR}) for *Semotilus atromaculatus* collected from sample sites in four catchments in the Kansas River basin. The dashed line (- - -) indicates the best-fit linear regression model for these data.

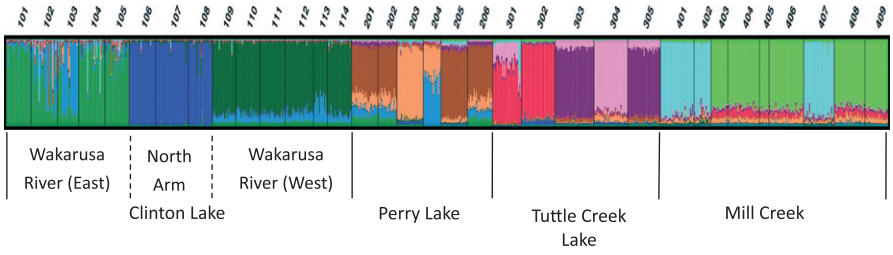

Fig. 2 Structure results showing eleven (K = 11) genetic clusters. Each vertical bar represents an individual grouped by sampling site. Coloured vertical bars indicate the proportional assignment of each individual multi-locus genotype to each genetic cluster.

Creek Lake were partitioned into three clusters corresponding to two sites separated by lotic habitat north of the impoundment, a single site east of the impoundment and two sites separated by lentic habitat (one east and one west of the impoundment). Sample sites in the Mill Creek catchment were partitioned into two genetic clusters (Figs 1 and 2).

Finally, pairwise genetic distance and geographical distance show a significant correlation across the study site (Mantel Test; $r = 0.466$, $P < 0.0001$), suggesting that genetic differentiation between creek chub populations is mediated, to some extent, by distance-limited dispersal (Fig. 3). Overall, these results suggest that catchments are largely independent genetic units subject to local regulation, justifying individual analysis for each.

Resistance to movement associated with impoundments

Genetic isolation was significantly influenced by an interaction between waterway distance and intervening habitat type (nonparametric ANCOVA, $P < 0.0001$). Significant patterns of isolation by distance were observed for populations separated primarily by lotic habitats (Fig. 4a, Mantel Test, $r = 0.369$, $P < 0.0001$) and those occurring in separate catchments (Fig. 4d, Mantel Test, $r = 0.245$, $P < 0.0001$). Populations separated primarily by lentic habitats (Fig. 4c, Mantel Test, $r = -0.427$, $P > 0.999$) or by

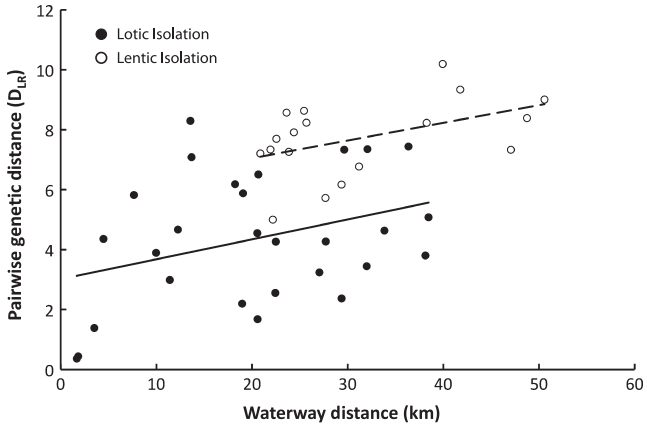

Fig. 5 Pairwise genetic distance (D_{LR}) as a function of pairwise waterway distance (km) for sample sites separated by lotic habitat (solid symbols) and sample sites separated by lentic habitat (open symbols) in the Clinton Lake catchment. The dashed and solid lines represent best-fit regression models for the lentic and lotic isolated sites, respectively.

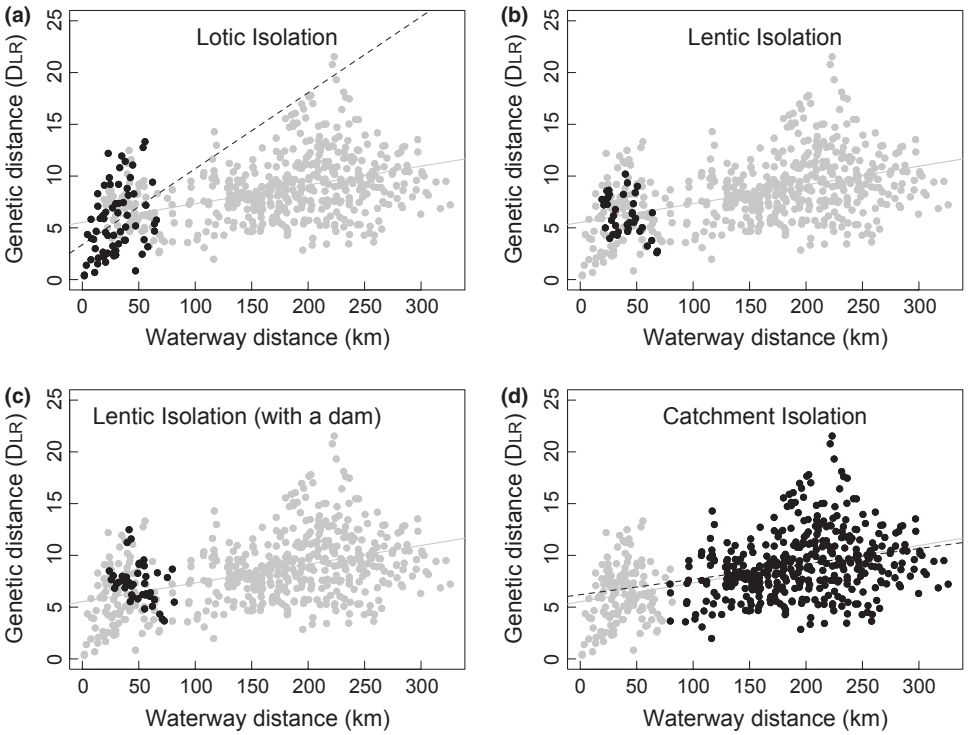

Fig. 4 Relationship between pairwise waterway distance (km) and pairwise genetic distance (D_{LR}) for *Semotilus atromaculatus* collected from sample sites (a) separated by lotic habitats, (b) separated by lentic habitats, (c) separated by lotic habitat, lentic habitat and a dam, and (d) contained within separate catchments. In each panel, the solid grey line (—) represents the best-fit linear regression of pairwise genetic distance on pairwise waterway distance for all sample sites (all points). The presence of a dashed black line (- - -) indicates significant isolation by distance and represents the best-fit linear regression for the subset of the samples considered in that panel (black points).

Table 4 BayesAss v 3.0 estimates of the mean proportion (±1 SD) of migrant individuals in each catchment classified by the catchment section (Fig. 1) and type of habitat separating population pairs included in the estimate. The proportion of migrant ancestry was estimated to be greatest in populations separated by lotic habitat, intermediate for population pairs separated by lentic habitat only and least for population pairs separated by lentic habitat and a dam. Only populations in the north branch of the Clinton Lake reservoir are estimated to exchange migrants at a statistically significant rate (bolded), based on 95% credible limits

		Intervening habitat type		
Catchment	Section	Lotic	Lentic	Lentic & Dam
Mill Creek		0.192 (0.130)		
Clinton Lake	West	0.024 (0.019)		
	North	**0.074 (0.014)**		
	Wakarusa	0.034 (0.021)		
	West & North		0.011 (0.011)	
	West/North & Wakarusa			0.013 (0.012)
Perry Lake		0.030 (0.025)		
			0.040 (0.023)	
Tuttle Creek Lake			0.035 (0.022)	
Average*		0.070 (0.042)	0.029 (0.019)	0.013 (0.012)
		0.040 (0.020)		

*The average proportion of migrant individuals is calculated including and excluding the estimate from Mill Creek.

lentic habitat and a dam (Fig. 4b, Mantel Test, $r = -0.362$, $P > 0.996$), however, did not follow a pattern of isolation by distance. Nonparametric ANCOVA revealed that intervening habitat type significantly affected genetic differentiation between populations in Clinton Lake ($P < 0.0001$) over a set of comparable distances (0–50 km). That is, mean pairwise genetic distance was greater in populations separated by lentic habitat compared to those separated by lotic habitat (Fig. 5). Moreover, average estimates of recent movement between populations (Table 4) calculated using BayesAss v3.0 were qualitatively greater among sites separated by lotic habitats (0.070 ± 0.042; mean ± 1 SD) than among sites separated by lentic habitat (0.029 ± 0.019) and lentic habitats and a dam (0.013 ± 0.012). Together, these results suggest that separation by lentic habitat reduces the connectivity between populations and therefore influences spatial genetic structure at a small scale.

Discussion

We used microsatellite data to assess creek chub genetic structure at 34 sample sites comprising a network of tributaries distributed across three impounded catchments and one un-impounded catchment in the Kansas River basin. Our analyses revealed that allelic richness and gene diversity estimates were consistent with large populations regardless of the location of the sampled tributaries with respect to an impoundment, and measures of heterozygote excess provided little evidence of recent population reductions. Nevertheless, our data indicate a high degree of spatial structuring, suggesting that the sample sites within these catchments comprise a set of isolated genetic units that are associated with catchments and subdivided within catchments into groups largely defined by intervening habitat type. Finally, when we considered dispersal with respect to intervening habitat type, we found that movement through lentic habitats of reservoirs increases resistance to dispersal among sites within a catchment.

In prairie streams, impoundment is associated with decreased abundance or localised extirpation in tributaries that are connected directly to reservoirs (Falke & Gido, 2006), and reduced abundance can result in the disruption of key reproductive and dispersal dynamics. For example, population growth rates for species that spawn semi-buoyant eggs (e.g. *Macrhybopsis* spp, *Hybognatus* spp) are partially contingent upon those eggs drifting downstream to develop in suitable habitat (Platania & Altenbach, 1998). In tributaries upstream of reservoirs, however, such eggs have a higher probability of drifting into unsuitable lake habitat and dying (Winston *et al.*, 1991), thus disrupting population dynamics and decreasing the stability of the population. This effect may be mitigated to some extent, however, if individuals from other, larger tributaries with more stable population dynamics are able to successfully disperse into, and thus rescue, populations destabilised by the presence of an impoundment (Campbell Grant *et al.*, 2007; Pulliam, 1988; Driscoll, 2007). Indeed, these sorts of population dynamics have been observed around impoundments for a common cyprinid that is a habitat generalist (*Cyprinella lutrensis*; Matthews & Marsh-Matthews, 2007). More commonly, however, genetic data from abundant, generalist stream-

dwelling fishes suggest the presence of an impoundment significantly disrupts connectivity among populations in tributaries (*sensu* Lowe, Likens & Power, 2006) as a result of newly created lentic habitat replacing a lotic migration corridor (e.g. Skalski *et al.*, 2008; Franssen, 2012).

Analysis of molecular variance and Bayesian clustering both revealed that the creek chub populations are organised into groups associated with catchments, and within a catchment, the genetic structure is largely influenced by the presence of an impoundment. Notably, however, the sample sites separated by lotic habitat and located along the Wakarusa River (East) downstream of the dam were subdivided within two clusters in the Clinton Lake system. Similarly, Mill Creek sites were subdivided into two genetic clusters. In both cases, the signal of structure may reflect barriers to gene flow independent of impoundments. Alternatively, isolation-by-distance patterns that represent more continuous genetic changes as distance between sites increases might have been treated as distinct breaks by the clustering algorithms in STRUCTURE (Pritchard *et al.*, 2000). The data from the Wakarusa River (East) indicate a high level of shared ancestry (relative to other clusters in our data) for individuals assigned to the two clusters. The close spatial positioning of the clustered sites coupled with the signal of mixed ancestry suggests that the clustering is an artefact produced by distance-limited dispersal between sites combined with signal contributed by un-sampled tributaries, both of which are known to influence the algorithm implemented in STRUCTURE (Pritchard *et al.*, 2000). For the Mill Creek drainage, the clustering of sites both low and high in the catchment is inconsistent with the presence of barriers and distance-limited dispersal as a structuring mechanism. Although fine-scale or more intensive sampling of these un-impounded stream networks might be necessary for fine-scale resolution of genetic structure, our data were sufficient to identify clear breaks in genetic structure associated with impoundments.

For creek chub in the Kansas River basin, genetic connectivity among the sample sites appears to be mediated, at least to some extent, by distance-limited dispersal as revealed by the pattern of isolation by distance at the among-catchments scale. Within catchments, however, the pattern of isolation by distance is absent, which implies one of three scenarios, as suggested by Hutchison & Templeton (1999): panmixia (Case II), differentiation driven by drift (Case III) or differentiation governed by the relative contributions of distance-limited dispersal and genetic drift (Case IV). Genotypes of individuals sampled from populations separated by lentic habitat or lentic habitats and a dam were, on average, 8.45 and 9.56 times more likely to be drawn from the allele pool at the sample site than the allele pool at the other site. We believe that this level of differentiation is more consistent with either Case III or Case IV of Hutchison & Templeton (1999), as one would expect more homogenous allele frequencies and smaller genetic distances between population pairs if the data were consistent with Case II. Discriminating between Case III and Case IV when considering the type of intervening habitat regardless of catchment, however, is more difficult. Isolation by distance was not apparent for populations separated by lentic habitats or lentic habitats and a dam, suggesting differentiation driven by genetic drift (Case III). It is important to note, however, that over a spatial scale of roughly 25–75 km the mean level of genetic differentiation between populations in both separation categories was similar to the mean pairwise differentiation between populations separated by lentic habitat. Hence, a lack of data from sites that are separated by <25 km prevents discrimination between Cases III and IV at this scale. Within the Wakarusa River catchment, however, the level of genetic isolation with respect to waterway distance is similar for sample sites separated by lotic habitat and those separated by lentic habitat, yet the mean level of differentiation for lentic-separated sites is significantly greater than that for lotic-separated sites. This pattern suggests a stronger role of drift compared to distance-limited dispersal when considering the forces that might influence differentiation between populations around Clinton Lake when compared to those in the Wakarusa River. Finally, average estimates of recent migration suggest that impoundments impose strong resistance to movement, reducing connectivity by approximately 27% when only lentic habitat intervenes between populations. Connectivity is further reduced by 55% when a dam disrupts the migration corridor in addition to lentic habitat. That is, waterway impoundment disrupts connectivity and dispersal dynamics, resulting in population differentiation driven primarily by genetic drift, but our data do not preclude connectivity between tributaries after impoundment.

Our results from comparisons of sites without intervening lentic habitat suggest dispersal dynamics consistent with the network dispersal model of Thornbrugh & Gido (2010). Briefly, the network dispersal model posits that individuals will periodically disperse from a tributary network, through a main stem (i.e. the Kansas River) and into a different tributary network, resulting in a pattern of genetic differentiation consistent with a stepping stone model and isolation by distance. This notion can be scaled

down to describe the pattern of dispersal within an isolated tributary network as well. In this case, the main stem is the tributary flowing directly into the Kansas River (e.g. the Mill Creek catchment) with the smaller tributaries forming the network. Regardless of scale, dispersal appears to be disrupted, but probably not completely severed, for populations separated by lentic habitats. Lentic habitat, therefore, appears to present a matrix that is resistant to dispersal of creek chub causing a transition to a pattern of dispersal consistent with the limited exchange model of Thornbrugh & Gido (2010), where tributaries and tributary networks are thought to rarely if ever exchange individuals.

Although genetic structure was clearly influenced by the presence of impoundments, our data revealed no erosion of genetic diversity in sample sites directly connected to or surrounding an impoundment when compared to sample sites in un-impounded systems. Detecting such an effect, however, is predicated on the notion that impoundment *per se* causes a reduction in population size (N_e) sufficient enough to induce a bottleneck and erode genetic diversity via drift. Although impoundment has been correlated with reduced gene diversity (H_E) in creek chub (Skalski *et al.*, 2008), the effect is not consistently observed in stream fish populations influenced by impoundment (e.g. Franssen, 2012; this study). Although it is possible to observe a high level of genetic diversity in recently bottlenecked populations because of a time lag between disturbance and the erosion of genetic diversity (Nei *et al.*, 1975; Cornuet & Luikart, 1996), this possibility seems unlikely given the lack of consistent heterozygote excess across loci within tributaries connected to impoundments. Thus, maintenance of genetic diversity in the Kansas River basin, where connectivity is clearly disrupted, suggests that catchment areas upstream of the impoundments are sufficient to support large effective populations above a reservoir, and these populations may serve as refuge from the negative impacts arising from community turnover in tributary reaches close to an impoundment or hydrological shifts as a result of drought (e.g. Falke & Gido, 2006; Matthews & Marsh-Matthews, 2007). If, however, upstream catchments are insufficiently large to buffer these negative effects, then the reduction of habitat area may result in the erosion of genetic diversity, and further, localised fragmentation may occur in times of mild or severe drought, resulting in drastic reductions in populations size and localised extirpation.

The picture that emerges from our data suggests that the presence of an impoundment can reduce the opportunity for dispersal and genetic rescue of populations inhabiting tributaries directly connected to an impoundment. Further, disruption of downstream habitats coupled with upstream dispersal bias (e.g. Skalski & Gilliam, 2000) may have an additive effect on isolation, compounding the deleterious effects that manifest in populations affected by impoundment. These patterns of isolation, however, may not be immediately deleterious to populations of locally abundant fishes if the size of the isolated habitat is large. It should be noted, however, that the bulk of evidence for the genetic effects of impoundment on stream fishes emerges from studies of species with high reproductive capacity, which are widespread and locally abundant. The negative effects observed in these species will likely be more pronounced in habitat specialists and benthic species as they often demonstrate high levels of spatial genetic structure even in relatively undisturbed reaches (Lamphere & Blum, 2012; Sterling *et al.*, 2012). Future research on the consequences of fragmentation on the viability of native stream fishes should seek to verify these results with other species, particularly those with different ecological traits. Such an analysis will help identify the characteristics of species that would make them vulnerable to extirpation when faced with fragmentation.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1. Multiplex sets, locus names with fluorescent labels (F = 6-FAM, N = NED, P = PET, V = VIC), primer concentrations, annealing temperatures (T_A), and numbers of polymerase chain reaction (PCR) cycles for 16 loci in creek chub.

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